Scientific Agent Catalogue

Boltzmann Labs science-environments • generated from repository contracts and task cards

Purpose

This catalogue lists declared specialist agents and summarizes their intended responsibility. An entry documents an agent contract; it is not by itself evidence that a production engine calculation has been qualified.

SPE (18 agents)

ID	Agent	Purpose
SPE-01	Spe 01 Small Molecule Spe	Compute and validate a single-point energy for a neutral small molecule. Use the [environment specification](../ENVIRONMENT_SPEC.md) for the scientific contract. Do not invent missing inputs, thresholds, engine capabilities or successful results.
SPE-02	Spe 02 Drug Like Molecule Spe	Compute a reproducible energy for a drug-like molecular system. Use the [environment specification](../ENVIRONMENT_SPEC.md) for the scientific contract. Do not invent missing inputs, thresholds, engine capabilities or successful results.
SPE-03	Spe 03 Conformer Energy Comparison	Compare conformer energies at one consistent theory level. Use the [environment specification](../ENVIRONMENT_SPEC.md) for the scientific contract. Do not invent missing inputs, thresholds, engine capabilities or successful results.
SPE-04	Spe 04 Charged Molecule Ion Spe	Evaluate a charged molecule without silently changing its state. Use the [environment specification](../ENVIRONMENT_SPEC.md) for the scientific contract. Do not invent missing inputs, thresholds, engine capabilities or successful results.
SPE-05	Spe 05 Open Shell Radical Spe	Evaluate an open-shell system with explicit spin treatment. Use the [environment specification](../ENVIRONMENT_SPEC.md) for the scientific contract. Do not invent missing inputs, thresholds, engine capabilities or successful results.
SPE-06	Spe 06 Dft Method Comparison	Compare methods only under explicitly compatible protocols. Use the [environment specification](../ENVIRONMENT_SPEC.md) for the scientific contract. Do not invent missing inputs, thresholds, engine capabilities or successful results.
SPE-07	Spe 07 Basis Set Convergence	Measure sensitivity to a frozen sequence of basis sets. Use the [environment specification](../ENVIRONMENT_SPEC.md) for the scientific contract. Do not invent missing inputs, thresholds, engine capabilities or successful results.
SPE-08	Spe 08 Electronic State Orbital Analysis	Inspect occupations and orbital evidence for the requested state. Use the [environment specification](../ENVIRONMENT_SPEC.md) for the scientific contract. Do not invent missing inputs, thresholds, engine capabilities or successful results.
SPE-09	Spe 09 Electron Density Electrostatic Property	Generate and validate density/ESP properties when supported. Use the [environment specification](../ENVIRONMENT_SPEC.md) for the scientific contract. Do not invent missing inputs, thresholds, engine capabilities or successful results.
SPE-10	Spe 10 Dispersion Noncovalent Interaction Spe	Assess a noncovalent interaction with compatible dispersion treatment. Use the [environment specification](../ENVIRONMENT_SPEC.md) for the scientific contract. Do not invent missing inputs, thresholds, engine capabilities or successful results.
SPE-11	Spe 11 Hydrogen Bonded Cluster Spe	Evaluate a hydrogen-bonded cluster without dropping monomers. Use the [environment specification](../ENVIRONMENT_SPEC.md) for the scientific contract. Do not invent missing inputs, thresholds, engine capabilities or successful results.
SPE-12	Spe 12 Metal Ligand Coordination Spe	Evaluate a supported metal-ligand complex with explicit state assumptions. Use the [environment specification](../ENVIRONMENT_SPEC.md) for the scientific contract. Do not invent missing inputs, thresholds, engine capabilities or successful results.
SPE-14	Spe 14 Reaction Energy Intermediate Spe	Evaluate a reaction intermediate against compatible reference states. Use the [environment specification](../ENVIRONMENT_SPEC.md) for the scientific contract. Do not invent missing inputs, thresholds, engine capabilities or successful results.
SPE-15	Spe 15 Qmmm Single Point Energy	Evaluate a prepared QM/MM partition and preserve boundary provenance. Use the [environment specification](../ENVIRONMENT_SPEC.md) for the scientific contract. Do not invent missing inputs, thresholds, engine capabilities or successful results.
SPE-17	Spe 17 Solvation Cluster Spe	Evaluate a solvation cluster with explicit solvent composition. Use the [environment specification](../ENVIRONMENT_SPEC.md) for the scientific contract. Do not invent missing inputs, thresholds, engine capabilities or successful results.
SPE-18	Spe 18 Molecular Strain Energy	Compute a strain energy from matched relaxed and constrained states. Use the [environment specification](../ENVIRONMENT_SPEC.md) for the scientific contract. Do not invent missing inputs, thresholds, engine capabilities or successful results.
SPE-19	Spe 19 Mlip Reference Energy	Compare an MLIP energy with a versioned QM reference set. Use the [environment specification](../ENVIRONMENT_SPEC.md) for the scientific contract. Do not invent missing inputs, thresholds, engine capabilities or successful results.
SPE-20	Spe 20 Spe Quality Control Recovery	Audit outputs, recover permitted failures and preserve provenance. Use the [environment specification](../ENVIRONMENT_SPEC.md) for the scientific contract. Do not invent missing inputs, thresholds, engine capabilities or successful results.

Geometry Optimisation (18 agents)

ID	Agent	Purpose
GEO-01	Geo Opt 01 Structure Input Validation	Validate molecular files, coordinates, elements and atom counts. Use the [environment specification](../ENVIRONMENT_SPEC.md) for the scientific contract. Do not invent missing inputs, thresholds, engine capabilities or successful results.
GEO-02	Geo Opt 02 Molecular Connectivity	Check bonds, valence and bond orders without inventing connectivity. Use the [environment specification](../ENVIRONMENT_SPEC.md) for the scientific contract. Do not invent missing inputs, thresholds, engine capabilities or successful results.
GEO-03	Geo Opt 03 Initial Conformation	Diagnose distorted geometries and severe starting clashes. Use the [environment specification](../ENVIRONMENT_SPEC.md) for the scientific contract. Do not invent missing inputs, thresholds, engine capabilities or successful results.
GEO-04	Geo Opt 04 Charge Multiplicity	Validate charge, electron count and multiplicity consistency. Use the [environment specification](../ENVIRONMENT_SPEC.md) for the scientific contract. Do not invent missing inputs, thresholds, engine capabilities or successful results.
GEO-05	Geo Opt 05 Qm Method Selection	Select only a supported method and basis for the declared system. Use the [environment specification](../ENVIRONMENT_SPEC.md) for the scientific contract. Do not invent missing inputs, thresholds, engine capabilities or successful results.
GEO-06	Geo Opt 06 Optimizer Selection	Choose a supported optimisation algorithm with rationale. Use the [environment specification](../ENVIRONMENT_SPEC.md) for the scientific contract. Do not invent missing inputs, thresholds, engine capabilities or successful results.
GEO-07	Geo Opt 07 Internal Coordinate Optimization	Assess Cartesian versus internal-coordinate optimisation. Use the [environment specification](../ENVIRONMENT_SPEC.md) for the scientific contract. Do not invent missing inputs, thresholds, engine capabilities or successful results.
GEO-08	Geo Opt 08 Constraints Boundaries	Validate fixed atoms, restraints, distance constraints and boundaries. Use the [environment specification](../ENVIRONMENT_SPEC.md) for the scientific contract. Do not invent missing inputs, thresholds, engine capabilities or successful results.
GEO-09	Geo Opt 09 Scf Convergence	Diagnose electronic convergence failure and propose a bounded retry. Use the [environment specification](../ENVIRONMENT_SPEC.md) for the scientific contract. Do not invent missing inputs, thresholds, engine capabilities or successful results.
GEO-10	Geo Opt 10 Geometry Convergence	Classify converged, unconverged and inconclusive optimisation logs. Use the [environment specification](../ENVIRONMENT_SPEC.md) for the scientific contract. Do not invent missing inputs, thresholds, engine capabilities or successful results.
GEO-11	Geo Opt 11 Optimization Stagnation	Detect oscillation, repeated geometries and stalled progress. Use the [environment specification](../ENVIRONMENT_SPEC.md) for the scientific contract. Do not invent missing inputs, thresholds, engine capabilities or successful results.
GEO-12	Geo Opt 12 Gradient Force Validation	Validate gradient shape, finiteness and force conventions. Use the [environment specification](../ENVIRONMENT_SPEC.md) for the scientific contract. Do not invent missing inputs, thresholds, engine capabilities or successful results.
GEO-13	Geo Opt 13 Hessian Minimum Verification	Distinguish a minimum from a saddle using supplied frequencies. Use the [environment specification](../ENVIRONMENT_SPEC.md) for the scientific contract. Do not invent missing inputs, thresholds, engine capabilities or successful results.
GEO-14	Geo Opt 14 Multi Conformer Optimization	Optimise and compare mapped conformers at one theory level. Use the [environment specification](../ENVIRONMENT_SPEC.md) for the scientific contract. Do not invent missing inputs, thresholds, engine capabilities or successful results.
GEO-15	Geo Opt 15 Protein Ligand Geometry	Assess clashes and coordination without treating every close contact as an error. Use the [environment specification](../ENVIRONMENT_SPEC.md) for the scientific contract. Do not invent missing inputs, thresholds, engine capabilities or successful results.
GEO-16	Geo Opt 16 Runtime Resource Recovery	Recover safely from timeout or memory events without changing science settings. Use the [environment specification](../ENVIRONMENT_SPEC.md) for the scientific contract. Do not invent missing inputs, thresholds, engine capabilities or successful results.
GEO-17	Geo Opt 17 Optimized Structure Quality	Independently validate final geometry, energy, gradients and metadata. Use the [environment specification](../ENVIRONMENT_SPEC.md) for the scientific contract. Do not invent missing inputs, thresholds, engine capabilities or successful results.
GEO-18	Geo Opt 18 Geo Opt Orchestrator	Route cases to relevant specialists and produce an auditable report. Use the [environment specification](../ENVIRONMENT_SPEC.md) for the scientific contract. Do not invent missing inputs, thresholds, engine capabilities or successful results.

HOMO–LUMO (18 agents)

ID	Agent	Purpose
HL-01	Homo Lumo 01 Molecular Input Validation	Validate supported structure formats and reject malformed or unsupported inputs. Use the [environment specification](../ENVIRONMENT_SPEC.md) for the scientific contract. Do not invent missing inputs, thresholds, engine capabilities or successful results.
HL-02	Homo Lumo 02 Molecular Connectivity	Check bonds, aromaticity, valence and hydrogen representation. Use the [environment specification](../ENVIRONMENT_SPEC.md) for the scientific contract. Do not invent missing inputs, thresholds, engine capabilities or successful results.
HL-03	Homo Lumo 03 Conformer Generation	Generate, deduplicate and select starting conformers. Use the [environment specification](../ENVIRONMENT_SPEC.md) for the scientific contract. Do not invent missing inputs, thresholds, engine capabilities or successful results.
HL-04	Homo Lumo 04 Geometry Preparation	Detect severe clashes and unsuitable geometries before QM execution. Use the [environment specification](../ENVIRONMENT_SPEC.md) for the scientific contract. Do not invent missing inputs, thresholds, engine capabilities or successful results.
HL-05	Homo Lumo 05 Charge And Spin	Validate charge, electron count and multiplicity assumptions. Use the [environment specification](../ENVIRONMENT_SPEC.md) for the scientific contract. Do not invent missing inputs, thresholds, engine capabilities or successful results.
HL-06	Homo Lumo 06 Qm Method Selection	Select a supported method and basis for the chemistry and budget. Use the [environment specification](../ENVIRONMENT_SPEC.md) for the scientific contract. Do not invent missing inputs, thresholds, engine capabilities or successful results.
HL-07	Homo Lumo 07 Basis Set Model Validation	Check basis availability, ECP requirements and reference compatibility. Use the [environment specification](../ENVIRONMENT_SPEC.md) for the scientific contract. Do not invent missing inputs, thresholds, engine capabilities or successful results.
HL-08	Homo Lumo 08 Scf Convergence	Diagnose SCF failure and apply only permitted recovery. Use the [environment specification](../ENVIRONMENT_SPEC.md) for the scientific contract. Do not invent missing inputs, thresholds, engine capabilities or successful results.
HL-09	Homo Lumo 09 Molecular Orbital Identification	Identify HOMO and LUMO indices from occupations and eigenvalues. Use the [environment specification](../ENVIRONMENT_SPEC.md) for the scientific contract. Do not invent missing inputs, thresholds, engine capabilities or successful results.
HL-10	Homo Lumo 10 Homo Lumo Gap Calculation	Compute and independently verify the orbital-energy gap and units. Use the [environment specification](../ENVIRONMENT_SPEC.md) for the scientific contract. Do not invent missing inputs, thresholds, engine capabilities or successful results.
HL-11	Homo Lumo 11 Orbital Visualization	Verify orbital plots correspond to reported indices. Use the [environment specification](../ENVIRONMENT_SPEC.md) for the scientific contract. Do not invent missing inputs, thresholds, engine capabilities or successful results.
HL-12	Homo Lumo 12 Orbital Character Analysis	Analyse orbital composition and localisation when data are available. Use the [environment specification](../ENVIRONMENT_SPEC.md) for the scientific contract. Do not invent missing inputs, thresholds, engine capabilities or successful results.
HL-13	Homo Lumo 13 Electronic State Stability	Check occupations, near-degeneracy and state-stability warnings. Use the [environment specification](../ENVIRONMENT_SPEC.md) for the scientific contract. Do not invent missing inputs, thresholds, engine capabilities or successful results.
HL-14	Homo Lumo 14 Multi Conformer Comparison	Compare gaps across conformers using one consistent protocol. Use the [environment specification](../ENVIRONMENT_SPEC.md) for the scientific contract. Do not invent missing inputs, thresholds, engine capabilities or successful results.
HL-15	Homo Lumo 15 Reference Accuracy Evaluation	Compare results with trusted references and declared tolerances. Use the [environment specification](../ENVIRONMENT_SPEC.md) for the scientific contract. Do not invent missing inputs, thresholds, engine capabilities or successful results.
HL-16	Homo Lumo 16 Runtime Resource Recovery	Monitor and recover jobs while preserving provenance. Use the [environment specification](../ENVIRONMENT_SPEC.md) for the scientific contract. Do not invent missing inputs, thresholds, engine capabilities or successful results.
HL-17	Homo Lumo 17 Result Validation Reporting	Check completeness, units, convergence and report consistency. Use the [environment specification](../ENVIRONMENT_SPEC.md) for the scientific contract. Do not invent missing inputs, thresholds, engine capabilities or successful results.
HL-18	Homo Lumo 18 Homo Lumo Orchestrator	Route dependencies, enforce budgets and package auditable evidence. Use the [environment specification](../ENVIRONMENT_SPEC.md) for the scientific contract. Do not invent missing inputs, thresholds, engine capabilities or successful results.

QM/MM (7 agents)

ID	Agent	Purpose
QMMM-PS01	QM Region Selection Auditor	Important catalytic residues, ligands, or cofactors omitted
QMMM-PS02	Boundary and Link-Atom Repair	Invalid boundary or link atoms
QMMM-PS03	Charge and Embedding Consistency	Inconsistent charge or electrostatic embedding
QMMM-PS04	SCF-in-QM/MM Recovery	QM-region SCF failure
QMMM-PS05	Coupling and Force Consistency	Energy/force coupling inconsistency
QMMM-PS06	QM Region Sensitivity Analyzer	Result depends on QM-region definition
QMMM-PS07	QM/MM Trajectory Failure Recovery	Coupled trajectory instability or failure

Molecular Dynamics (62 agents)

ID	Agent	Purpose
MD-01	System Input Validation	Specialist responsibility: core
MD-02	System Preparation	Specialist responsibility: core
MD-03	Force-Field Selection	Specialist responsibility: core
MD-04	Parameterization	Specialist responsibility: core
MD-05	Solvation and Ion Placement	Specialist responsibility: core
MD-06	Energy Minimization	Specialist responsibility: core
MD-07	Equilibration	Specialist responsibility: core
MD-08	Integrator and Timestep	Specialist responsibility: core
MD-09	Thermostat and Barostat	Specialist responsibility: core
MD-10	Simulation Execution	Launch, monitor, checkpoint and resume a declared MD job. Use the [environment specification](../ENVIRONMENT_SPEC.md) for the shared contract. Do not invent trajectories, model suitability, benchmark scores or online-resource evidence.
MD-11	Failure Diagnosis and Recovery	Diagnose crashes, instabilities and failed jobs with bounded recovery. Use the [environment specification](../ENVIRONMENT_SPEC.md) for the shared contract. Do not invent trajectories, model suitability, benchmark scores or online-resource evidence.
MD-12	Sampling Strategy	Recommend length, replicas and sampling design for the scientific question. Use the [environment specification](../ENVIRONMENT_SPEC.md) for the shared contract. Do not invent trajectories, model suitability, benchmark scores or online-resource evidence.
MD-13	Enhanced Sampling	Configure and validate a supported metadynamics, replica-exchange or adaptive method. Use the [environment specification](../ENVIRONMENT_SPEC.md) for the shared contract. Do not invent trajectories, model suitability, benchmark scores or online-resource evidence.
MD-14	Trajectory Quality Control	Detect corruption, drift, missing frames and incomplete trajectories. Use the [environment specification](../ENVIRONMENT_SPEC.md) for the shared contract. Do not invent trajectories, model suitability, benchmark scores or online-resource evidence.
MD-15	Structural Analysis	Calculate and validate RMSD, RMSF, radius of gyration and clustering. Use the [environment specification](../ENVIRONMENT_SPEC.md) for the shared contract. Do not invent trajectories, model suitability, benchmark scores or online-resource evidence.
MD-16	Interaction Analysis	Analyse hydrogen bonds, contacts, salt bridges and coordination. Use the [environment specification](../ENVIRONMENT_SPEC.md) for the shared contract. Do not invent trajectories, model suitability, benchmark scores or online-resource evidence.
MD-17	Free-Energy Analysis	Execute or validate a supported free-energy workflow with uncertainty. Use the [environment specification](../ENVIRONMENT_SPEC.md) for the shared contract. Do not invent trajectories, model suitability, benchmark scores or online-resource evidence.
MD-18	Statistical and Uncertainty Analysis	Evaluate convergence, uncertainty and independent replica agreement. Use the [environment specification](../ENVIRONMENT_SPEC.md) for the shared contract. Do not invent trajectories, model suitability, benchmark scores or online-resource evidence.
MD-19	Resource Optimization	Manage GPU/CPU throughput, memory and compute budgets without changing science. Use the [environment specification](../ENVIRONMENT_SPEC.md) for the shared contract. Do not invent trajectories, model suitability, benchmark scores or online-resource evidence.
MD-20	MD Orchestrator and Audit	Route stages, enforce gates, coordinate specialists and package an audit trail. Use the [environment specification](../ENVIRONMENT_SPEC.md) for the shared contract. Do not invent trajectories, model suitability, benchmark scores or online-resource evidence.
MD-21	Online Scientific Literature	Discover papers, methods, benchmarks and limitations with citation provenance. Use the [environment specification](../ENVIRONMENT_SPEC.md) for the shared contract. Do not invent trajectories, model suitability, benchmark scores or online-resource evidence.
MD-22	Model and Checkpoint Discovery	Find candidate learned models/checkpoints and inspect usage constraints. Use the [environment specification](../ENVIRONMENT_SPEC.md) for the shared contract. Do not invent trajectories, model suitability, benchmark scores or online-resource evidence.
MD-23	Scientific Dataset Discovery	Find datasets, schemas, metadata, licenses and suitable subsets. Use the [environment specification](../ENVIRONMENT_SPEC.md) for the shared contract. Do not invent trajectories, model suitability, benchmark scores or online-resource evidence.

ID	Agent	Purpose
MD-24	Resource Download and Cache	Download approved resources with resumable transfers and immutable cache records. Use the [environment specification](../ENVIRONMENT_SPEC.md) for the shared contract. Do not invent trajectories, model suitability, benchmark scores or online-resource evidence.
MD-25	Model Compatibility	Check architecture, framework, hardware, units and supported molecular systems. Use the [environment specification](../ENVIRONMENT_SPEC.md) for the shared contract. Do not invent trajectories, model suitability, benchmark scores or online-resource evidence.
MD-26	Dataset Integrity	Verify hashes, formats, completeness, units and provenance of downloaded data. Use the [environment specification](../ENVIRONMENT_SPEC.md) for the shared contract. Do not invent trajectories, model suitability, benchmark scores or online-resource evidence.
MD-27	Scientific Reproduction	Reconstruct a published benchmark using its stated settings and evidence. Use the [environment specification](../ENVIRONMENT_SPEC.md) for the shared contract. Do not invent trajectories, model suitability, benchmark scores or online-resource evidence.
MD-28	Model Evaluation and Selection	Compare candidate models with a fixed baseline on held-out tests. Use the [environment specification](../ENVIRONMENT_SPEC.md) for the shared contract. Do not invent trajectories, model suitability, benchmark scores or online-resource evidence.
MD-29	Online Resource Update	Detect new versions and record changes without replacing validated resources silently. Use the [environment specification](../ENVIRONMENT_SPEC.md) for the shared contract. Do not invent trajectories, model suitability, benchmark scores or online-resource evidence.
MD-30	Provenance and Licence	Track authorship, license, source URL, version, citation and permitted usage. Use the [environment specification](../ENVIRONMENT_SPEC.md) for the shared contract. Do not invent trajectories, model suitability, benchmark scores or online-resource evidence.
MD-31	Protein Folding and Misfolding Specialist	Analyse folding pathways, conformational stability, misfolding intermediates and structural transitions. Use the [environment specification](../ENVIRONMENT_SPEC.md) for the shared contract. Do not invent trajectories, model suitability, benchmark scores or uncertainty.
MD-32	Protein–Protein Interaction MD Specialist	Evaluate interface stability, contact persistence, hydrogen bonds and interaction networks. Use the [environment specification](../ENVIRONMENT_SPEC.md) for the shared contract. Do not invent trajectories, model suitability, benchmark scores or uncertainty.
MD-33	Membrane Protein MD Specialist	Validate membrane embedding, lipid environment, protein orientation and compatibility. Use the [environment specification](../ENVIRONMENT_SPEC.md) for the shared contract. Do not invent trajectories, model suitability, benchmark scores or uncertainty.
MD-34	Nucleic Acid Dynamics Specialist	Analyse DNA/RNA base pairing, stacking, deviations and conformational states. Use the [environment specification](../ENVIRONMENT_SPEC.md) for the shared contract. Do not invent trajectories, model suitability, benchmark scores or uncertainty.
MD-35	Enzyme Mechanism MD Specialist	Investigate active-site dynamics, catalytic networks and substrate positioning. Use the [environment specification](../ENVIRONMENT_SPEC.md) for the shared contract. Do not invent trajectories, model suitability, benchmark scores or uncertainty.
MD-36	Binding-Site Water Thermodynamics Specialist	Study hydration sites, water residence, displacement and pocket networks. Use the [environment specification](../ENVIRONMENT_SPEC.md) for the shared contract. Do not invent trajectories, model suitability, benchmark scores or uncertainty.
MD-37	Conformational State Clustering Specialist	Identify metastable states using RMSD, contacts and dimensionality reduction. Use the [environment specification](../ENVIRONMENT_SPEC.md) for the shared contract. Do not invent trajectories, model suitability, benchmark scores or uncertainty.
MD-38	Markov State Model Specialist	Build and validate Markov state models with lag-time and implied-timescale checks. Use the [environment specification](../ENVIRONMENT_SPEC.md) for the shared contract. Do not invent trajectories, model suitability, benchmark scores or uncertainty.
MD-39	Transition Path Analysis Specialist	Analyse transitions between states and transition-path ensembles. Use the [environment specification](../ENVIRONMENT_SPEC.md) for the shared contract. Do not invent trajectories, model suitability, benchmark scores or uncertainty.
MD-40	Replica Exchange MD Specialist	Check replica assignments, exchanges, acceptance and sampling convergence. Use the [environment specification](../ENVIRONMENT_SPEC.md) for the shared contract. Do not invent trajectories, model suitability, benchmark scores or uncertainty.
MD-41	Potential of Mean Force Specialist	Estimate free-energy profiles along declared collective variables with uncertainty. Use the [environment specification](../ENVIRONMENT_SPEC.md) for the shared contract. Do not invent trajectories, model suitability, benchmark scores or uncertainty.

ID	Agent	Purpose
MD-42	Ligand Unbinding Pathway Specialist	Characterise ligand exit pathways, bottlenecks and residence-time evidence. Use the [environment specification](../ENVIRONMENT_SPEC.md) for the shared contract. Do not invent trajectories, model suitability, benchmark scores or uncertainty.
MD-43	Relative Binding Affinity Validation Specialist	Check FEP/TI cycles, overlap, convergence and uncertainty across ligand pairs. Use the [environment specification](../ENVIRONMENT_SPEC.md) for the shared contract. Do not invent trajectories, model suitability, benchmark scores or uncertainty.
MD-44	QM/MM Boundary and Region Specialist	Check QM-region selection, link atoms, boundary treatment and sensitivity plans. Use the [environment specification](../ENVIRONMENT_SPEC.md) for the shared contract. Do not invent trajectories, model suitability, benchmark scores or uncertainty.
MD-45	Machine-Learned Potential Validation Specialist	Evaluate ML potentials against reference energies, forces and extrapolation regions. Use the [environment specification](../ENVIRONMENT_SPEC.md) for the shared contract. Do not invent trajectories, model suitability, benchmark scores or uncertainty.
MD-46	Adaptive Sampling Controller	Select new simulations from under-sampled states and uncertainty. Use the [environment specification](../ENVIRONMENT_SPEC.md) for the shared contract. Do not invent trajectories, model suitability, benchmark scores or uncertainty.
MD-47	Trajectory Reproducibility Auditor	Audit inputs, seeds, engine versions, parameters, trajectory integrity and analysis provenance. Use the [environment specification](../ENVIRONMENT_SPEC.md) for the shared contract. Do not invent trajectories, model suitability, benchmark scores or uncertainty.
MD-48	Statistical Convergence and Uncertainty Specialist	Assess autocorrelation, effective sample size, block uncertainty and replica consistency. Use the [environment specification](../ENVIRONMENT_SPEC.md) for the shared contract. Do not invent trajectories, model suitability, benchmark scores or uncertainty.
MD-49	MD Benchmark Dataset Builder	Create versioned benchmark cases, splits, checksums and reference provenance. Use the [environment specification](../ENVIRONMENT_SPEC.md) for the shared contract. Do not invent trajectories, model suitability, benchmark scores or uncertainty.
MD-50	End-to-End MD Qualification Orchestrator	Coordinate specialists, verify evidence and apply qualification gates. Use the [environment specification](../ENVIRONMENT_SPEC.md) for the shared contract. Do not invent trajectories, model suitability, benchmark scores or uncertainty.
MD-51	Molecular System Preparation Scenarios	See the module task card for the declared scientific responsibility.
MD-52	Force-Field and Parameterization Failure Scenarios	See the module task card for the declared scientific responsibility.
MD-53	Minimization and Equilibration Failure Scenarios	See the module task card for the declared scientific responsibility.
MD-54	Production MD Execution Failure Scenarios	See the module task card for the declared scientific responsibility.
MD-55	Trajectory and Structural Analysis Scenarios	See the module task card for the declared scientific responsibility.
MD-56	Sampling and Convergence Failure Scenarios	See the module task card for the declared scientific responsibility.
MD-57	Drug Discovery and Binding Scenarios	See the module task card for the declared scientific responsibility.
MD-58	Machine-Learning Potential Scenarios	See the module task card for the declared scientific responsibility.
MD-59	Online Model and Dataset Retrieval Scenarios	See the module task card for the declared scientific responsibility.
MD-60	Compute and Resource Constraint Scenarios	See the module task card for the declared scientific responsibility.
MD-61	Scientific Validation and Reproducibility Scenarios	See the module task card for the declared scientific responsibility.
MD-62	Autonomous Agent and Orchestration Scenarios	See the module task card for the declared scientific responsibility.

Coarse-Grained MD (20 agents)

ID	Agent	Purpose
CG-01	CG Representation and Mapping Agent	Specialist responsibility: mapping
CG-02	CG Topology Builder Agent	Specialist responsibility: mapping
CG-03	CG Force-Field Selector Agent	Specialist responsibility: forcefield
CG-04	CG Parameterization Agent	Specialist responsibility: forcefield
CG-05	CG Model Transferability Agent	Specialist responsibility: validation
CG-06	CG System Builder Agent	Specialist responsibility: system_setup
CG-07	CG Energy Minimization Agent	Specialist responsibility: execution
CG-08	CG Equilibration Agent	Specialist responsibility: execution
CG-09	CG Simulation Stability Agent	Specialist responsibility: execution
CG-10	CG Sampling Strategy Agent	Validate RDF, density, radius of gyration, diffusion, phase behavior, convergence, trajectory integrity, replicas, and reference agreement. The specialists are Uncertainty/Sensitivity (`CG-19`) and Benchmark/Qualification (`CG-20`).
CG-11	CG Protein Structure Validation Agent	Specialist responsibility: analysis
CG-12	CG Membrane Validation Agent	Specialist responsibility: analysis
CG-13	CG Phase Separation Agent	Specialist responsibility: analysis
CG-14	CG Self-Assembly Agent	Specialist responsibility: analysis
CG-15	CG Structural Observable Matching Agent	Specialist responsibility: validation
CG-16	CG Thermodynamic Validation Agent	Specialist responsibility: validation
CG-17	CG Kinetics and Dynamics Validation Agent	Specialist responsibility: validation
CG-18	CG-to-Atomistic Reconstruction Agent	Specialist responsibility: multiscale
CG-19	CG Uncertainty and Sensitivity Agent	Specialist responsibility: validation
CG-20	CG Benchmark and Qualification Agent	Specialist responsibility: orchestration

FEP/TI specialist agents (15 agents)

ID	Agent	Purpose
FEP-AG01	Alchemical Mapping Auditor	Incorrect atom mapping
FEP-AG02	Transformation Feasibility Agent	Invalid transformation
FEP-AG03	Force-Field and Parameter Auditor	Missing or incompatible parameters
FEP-AG04	Lambda Schedule Optimizer	Unstable or poorly spaced windows
FEP-AG05	Charge, Restraint and Correction Auditor	Corrections or standard-state errors
FEP-AG06	Simulation Failure Diagnostician	Crashes, NaNs, failed replicas
FEP-AG07	Phase-Space Overlap Auditor	Poor neighboring-window overlap
FEP-AG08	Convergence and Hysteresis Auditor	Drift or replica disagreement
FEP-AG09	Binding-Site Water and Slow-State Investigator	Trapped water or slow pocket rearrangement
FEP-AG10	Adaptive Sampling Controller	Compute allocated to easy windows
FEP-AG11	Free-Energy Estimator Auditor	Estimator, sign, unit, or uncertainty error
FEP-AG12	Thermodynamic Cycle Auditor	Missing or inconsistent legs
FEP-AG13	Relative Affinity Ranking Auditor	Unsupported ranking conclusion
FEP-AG14	Scientific False-Success Auditor	Technical completion but invalid science
FEP-AG15	Independent Qualification Agent	Insufficient final evidence

FEP/TI problem-solving agents (10 agents)

ID	Agent	Purpose
FEP-PS01	Alchemical Mapping Auditor	Incorrect atom mapping between reference and target ligands
FEP-PS02	Transformation Feasibility Agent	Invalid or chemically inconsistent transformation
FEP-PS03	Lambda Schedule Optimizer	Poorly spaced lambda windows or unstable intermediates
FEP-PS04	Phase-Space Overlap Auditor	Insufficient neighboring-window overlap
FEP-PS05	Free-Energy Convergence Agent	Drift or replica disagreement
FEP-PS06	Hysteresis and Cycle-Closure Auditor	Forward/reverse disagreement or broken cycles
FEP-PS07	Statistical Uncertainty and Estimator Auditor	Invalid estimator or uncertainty
FEP-PS08	Charge-Changing Transformation Auditor	Electrostatic finite-size artifacts
FEP-PS09	Restraint and Standard-State Auditor	Biased restraints or standard-state terms
FEP-PS10	Free-Energy Result Qualification Agent	Number reported without reliability evidence

Total declared agents: 168
Source: versioned files under environments/*/agents and environments/*/tasks. Regenerate with scripts/build_agent_catalog_pdf.py.